

ASHUTOSH SOMVANSHI

Senior AI Engineer & GenAI Systems Architect

Email: ashutoshsomvanshi247@gmail.com | Phone: +91-6394256455 | Location: India (Open to 100% Remote)

LinkedIn: linkedin.com/in/ashutoshsomvanshi | GitHub: github.com/ashutosh-somvanshi

PROFESSIONAL SUMMARY

Senior AI Engineer & Solutions Architect with 5+ years of dedicated hands-on experience architecting and deploying enterprise-scale Generative AI, LLM systems, Agentic workflows, and production RAG platforms. Proven track record translating complex enterprise requirements into scalable, secure AI products serving 5,000+ enterprise users. Certified in both Anthropic Claude Architecture and Microsoft Azure AI, with deep expertise in Python, LangGraph, Semantic Kernel, Azure OpenAI, Vector Databases, and enterprise system integrations. Adept at bridging concept-to-production lifecycle, optimizing inference latency and token economics, and delivering high-ROI AI automation.

CORE TECHNICAL SKILLS

- Generative AI & LLMs:** LLM Orchestration, Multi-Agent Systems, Agentic Workflows, Production RAG, Contextual Memory, Prompt Optimization, Guardrails & Model Evaluation.
- Agent Frameworks & Tooling:** LangGraph, Semantic Kernel, LangChain, AutoGen, CrewAI, Model Context Protocol (MCP), Microsoft Bot Framework SDK.
- Model Ecosystem & Cloud AI:** Azure OpenAI (GPT-4o, o1/o3), Anthropic Claude (Claude 3.5 Sonnet/Opus), Open-Source LLMs (Llama 3, Mistral), AI Foundry, Azure AI Search, Azure Cognitive Services.
- Vector Search & Retrieval:** Hybrid Search (Dense + Sparse BM25), Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, Vector Databases (Azure AI Search, Qdrant, pgvector), Embedding Models.
- Engineering Stack & Backend:** Python (Expert), FastAPI, AsyncIO, REST APIs, GraphQL, SQL, Docker, CI/CD, Azure DevOps, Git.
- Enterprise Integration & Security:** Microsoft 365, Microsoft Graph API, Dynamics 365, SharePoint, OAuth 2.0, Enterprise RBAC, Private Endpoints, PII Masking, Token Cost Optimization.

PROFESSIONAL CERTIFICATIONS

- Claude Certified Architect: Foundation** — Anthropic (2025/2026)
- Microsoft Certified: Azure AI Engineer Associate** (2025)
- Oracle Cloud Infrastructure: Generative AI Professional** (2025)
- LangChain Academy:** Introduction to LangGraph & Agentic Workflows (2025)
- Hugging Face:** Fundamentals of AI Agents (2025)
- AWS + DeepLearning.AI:** Generative AI with Large Language Models (LLMs) (2024)

PROFESSIONAL EXPERIENCE

Senior AI Lead – Generative AI & Enterprise Architecture

Nihilent Ltd | September 2024 – Present

- Enterprise Copilot & Assistant Architecture:** Architected and delivered intelligent conversational copilots and custom GPT solutions integrated with Microsoft Teams, SharePoint, and Dynamics 365, serving 5,000+ enterprise users with a 95% resolution satisfaction rate.
- Production Hybrid RAG Systems:** Engineered high-accuracy RAG pipelines using Azure AI Search, dense embeddings, and cross-encoder reranking over 100,000+ complex enterprise documents, boosting context retrieval precision from 68% to 94%.
- Agentic Workflow Automation:** Designed stateful multi-agent workflows using LangGraph and Semantic Kernel, orchestrating tool-calling schemas, state checkpointing, and human-in-the-loop validation for automated enterprise business reporting.
- AI Lifecycle & FinOps Optimization:** Leveraged Azure AI Foundry and custom telemetry to optimize model selection, token budgets, and prompt compression, driving a 40% reduction in inference latency and significant cloud operational cost savings.
- Enterprise Data Ingestion:** Developed secure data connectors using Microsoft Graph API, REST, and GraphQL to ground LLM reasoning across disparate enterprise data silos with strict role-based access control (RBAC).
- Executive POC-to-Production:** Spearheaded 15+ AI proof-of-concepts (POCs) from initial scoping to production release, partnering with enterprise product owners to identify high-ROI automation use cases.

AI Technical Lead – Conversational AI & Enterprise Integration

Nihilent Ltd | July 2023 – August 2024

- **Multi-Turn Conversational Bots:** Developed complex multi-turn conversational agents with Microsoft Bot Framework SDK and Azure OpenAI APIs, implementing advanced state management and contextual memory retention.
- **Reusable Enterprise AI Accelerators:** Built reusable architectural templates and modular Python middleware for LLM rate-limiting, semantic caching, and error recovery, reducing enterprise AI feature delivery time by 60%.
- **CI/CD & MLOps Governance:** Collaborated with cloud DevOps teams to establish automated CI/CD deployment pipelines, prompt evaluation gates, and telemetry monitoring in Azure DevOps.
- **Technical Mentorship:** Mentored 10+ junior and mid-level engineers in LLM orchestration, prompt design best practices, and secure API development.

AI/ML Engineer – Machine Learning Solutions

Nihilent Ltd / Prior Technical Experience | 2019 – 2023

- **NLP & Predictive Modeling:** Built and tuned traditional ML and NLP pipelines using Python, Scikit-Learn, and BERT for text classification, sentiment analysis, and intent recognition.
- **Vector Search & Embeddings Exploration:** Implemented early semantic search prototypes using embedding models and vector indexing algorithms (HNSW, Cosine Similarity) to improve document discovery.
- **Cloud ML Deployments:** Configured and deployed model endpoints using Azure ML, Azure Cognitive Services, and containerized Docker microservices.

KEY STRATEGIC PORTFOLIO PROJECTS

- **Enterprise Cognitive RAG Platform (DocuMind AI):** Designed an enterprise knowledge retrieval engine combining Dense Vector Search and BM25 Sparse Search via Reciprocal Rank Fusion (RRF) and Cross-Encoder reranking. Integrated Redis semantic caching to deliver sub-500ms P99 latency and slash token costs by 42%. Evaluated via Ragas with 96.4% faithfulness.
- **Multi-Agent Autonomous Incident Triage Engine (IncidentOps AI):** Built a fault-tolerant multi-agent state machine in Python & LangGraph with Postgres state persistence. Orchestrated specialized diagnostic agents with tool-calling schemas and a Human-in-the-Loop Slack approval gate for automated cloud alert diagnosis.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science and Engineering

Vishveshwarya Institute of Engineering and Technology (VIET) | Graduated 2022